

LibraryJDBCSystem

Version control

LibraryJDBCSystem

LibraryJDBCSystem.java

Project

LibraryJDBCSystem

idea

out

src

BookNotAvailableException

Faculty

LibraryJDBCSystem

LibraryMember

MembershipExpiredException

Student

External Libraries

Scratches and Consoles

LibraryJDBCSystem.java

1 import java.sql.*;

2 import java.time.LocalDate;

3 import java.util.Arrays;

4

5 public class LibraryJDBCSystem {

6 private static final String URL = "jdbc:mysql://localhost:3306/Librarydb";

7 private static final String USER = "root";

8 private static final String PASSWORD = "root";

9

10 public static void main(String[] args) {

11 try {

12 Class.forName("com.mysql.cj.jdbc.Driver");

13

14 try (Connection con = DriverManager.getConnection(URL, USER, PASSWORD)) {

Run

LibraryJDBCSystem

C:\Program Files\Java\jdk-17\bin\java.exe -javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.1\lib\idea_rt.jar=62312:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.1\bin -Dfile.encoding=UTF-8 -classpath C:\Users\admin\IdeaProjects\LibraryJDBCSystem\out\production\LibraryJDBCSystem LibraryJDBCSystem

Database connected successfully.

Member ID: 101

Name: Arun

Email: null

Phone: null

Maximum Books Allowed: 3

Member ID: 201

Name: Dr. Kumar

Email: kumar@college.edu

Phone: 9876543210

Maximum Books Allowed: 10

--- Issuing Single Book ---

Book 1001 successfully issued to Arun

--- Issuing Multiple Books ---

All books issued successfully.

--- Transaction Test ---

AutoCommit = false

1. Updating books table...

UPDATE completed. Rows affected: 1

2. Inserting into issued_books...

INSERT failed.

Reason: Duplicate entry '101-1001' for key 'issued_books.member_id'

Executing ROLLBACK...

ROLLBACK successful.

The previous UPDATE has also been undone.

AutoCommit restored.

Process finished with exit code 0

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

librarydb

Tables

books

issued_books

Views

Stored Procedures

Functions

mysql

performance_schema

sys

Administration Schemas

Information

Table: issued_books

Columns:

issue_id int AI PK

member_id int

isbn int

issue_date date

Query 1

1 • SELECT * FROM librarydb.books;

Result Grid

isbn	title	author	copies
1001	Java Programming	James Gosling	4
1002	Data Structures	Mark Allen	2
1003	Database Systems	C. J. Date	1

Query 2

1 • SELECT * FROM librarydb.issued_books;

Result Grid

issue_id	member_id	isbn	issue_date
1	101	1001	2025-05-24
2	201	1002	2025-05-24
3	201	1003	2025-05-24

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:20:10	SELECT * FROM librarydb.books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
2	15:20:15	SELECT * FROM librarydb.issued_books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
3	15:21:02	UPDATE books SET copies = copies - 1 WHERE isbn = 1001 AND copies > 0	1 row(s) affected	0.016 sec
4	15:21:02	INSERT INTO issued_books(member_id, isbn, issue_date) VALUES (101, 1001, CURDATE())	1 row(s) affected	0.015 sec
5	15:21:20	ROLLBACK	0 row(s) affected	0.001 sec

Query Completed

SQL Additions

Jump to

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Run: LibraryDBSystem

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AutoCommit restored.

Process finished with exit code 0